

[Submission date]

Editor-in-Chief

Technological Forecasting and Social Change

Dear Editor,

We submit the full-length research article “Legacy standards as frozen solutions to obsolete optimization problems: a framework for technological change and adaptive re-optimization” for consideration in Technological Forecasting and Social Change.

The manuscript develops a pre-specified Legacy Standard Audit that separates historical, current frictionless, and transition-adjusted optima. Its contribution is methodological and governance-oriented: it makes claims of technology-driven mismatch falsifiable while preserving null, contrary, and insufficient-evidence outcomes.

The empirical demonstration began from 122 candidates in 12 domains and froze 7 heterogeneous cases before confirmatory estimation. Results are deliberately not pooled because objectives and units are incomparable. The manuscript makes no general causal claim and withholds redesign recommendations where switching, compatibility, safety, or coordination costs are not observed.

The work fits the journal because specific technological developments—sensing, computation, communication, prognostics, interval metering, and adaptive control—are the central mechanisms that alter feasible decision sets. The paper connects these mechanisms to standards governance, transition costs, and policy implications.

The manuscript is original, is not under consideration elsewhere, and uses public secondary data and simulations. All authors must confirm authorship, conflicts, and the final declarations before submission.

Sincerely,

[Corresponding author name and affiliation]